Why and How to Think with AI

Today you will create your own AI agent and use it to think.

timgasste.github.io/ai-thinking

Before we start, open this page on your laptop and follow **Get ready**: download ChatGPT, create an account, and set up your folder. More on that in the lecture.

Tim Gåsste
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Will AI Take Your Job?



Jensen Huang
CEO, Nvidia

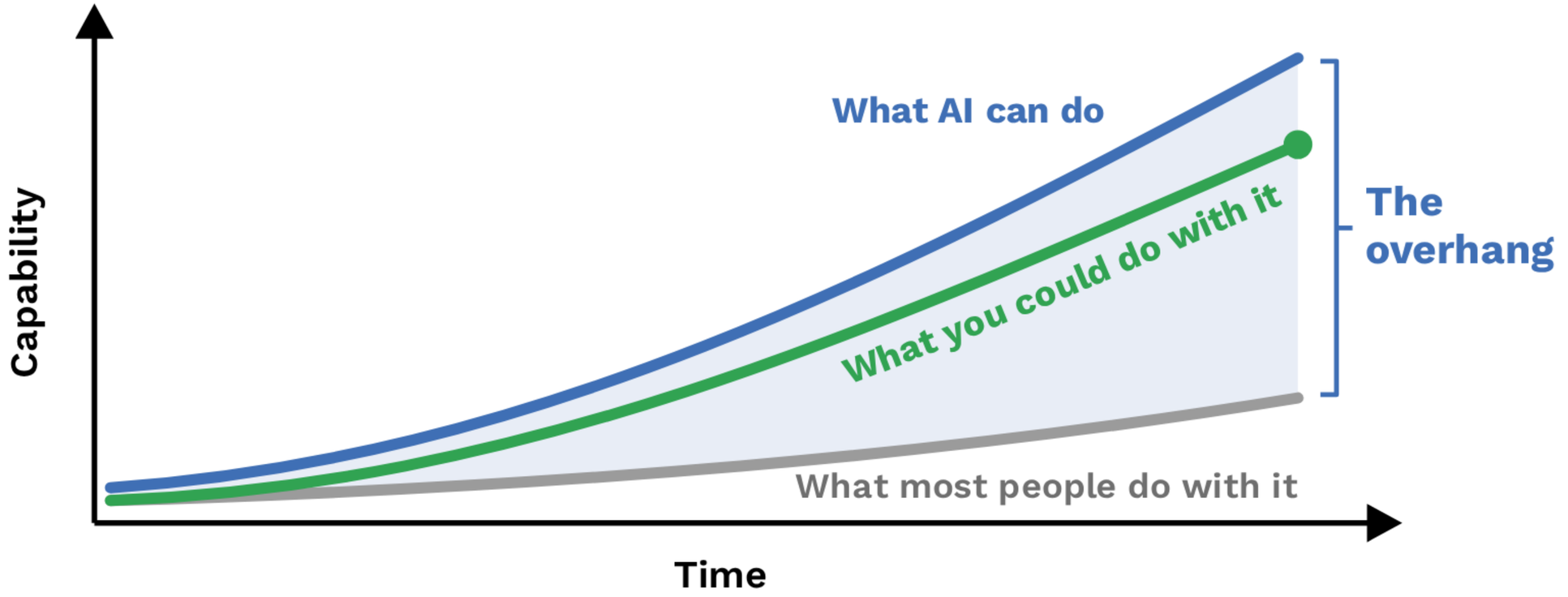
> You're not going to lose your job to an AI, but you're going to lose your job to **someone who uses AI**.

Milken Institute Global Conference, 6 May 2025, as reported by CNBC. Photo: The White House (public domain).

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The Overhang

Ethan Mollick calls it **the overhang**: “the gap between what these models can do and what almost anyone is doing with them.”

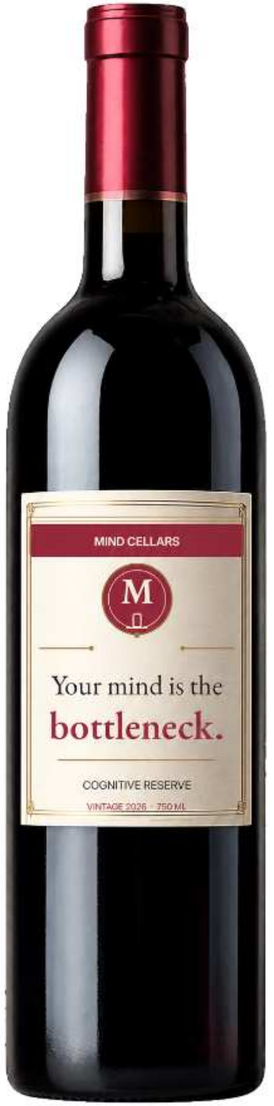


Conceptual illustration, not data. Quote: Ethan Mollick, “The Overhang,” *One Useful Thing*, 18 September 2026.

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Your Capacity to Think Is the Bottleneck

- **AI increasingly does the execution**: searching, drafting, coding, summarizing.
- What it multiplies is what you bring: **specialized and general knowledge, coupled with creativity, taste, and judgment.**
- All of these are built by **thinking and learning**. That is the bottleneck, and it is yours to widen.



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AI moves the hard thinking **upstream**:
into framing, judging, and deciding.
Without expertise, you get **polished slop**.



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Working at a higher level of abstraction does not make knowledge less important. **It does the opposite.**

CASE IN POINT · NPR, 7 JULY 2026

“Why AI companies are hiring philosophers to help develop their models”

Anthropic and Google bring in philosophers to help shape how their models reason and behave.

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Become a Unicorn

Four human advantages that AI multiplies

Deep knowledge
Expertise so thorough it becomes intuition. You sense when something is wrong.

Wide knowledge
Read across fields, formats, and traditions. It helps you understand and ask the right questions.

Taste
AI makes output abundant. Choosing well is the scarce skill.

Agency
Explore what AI can do in your field. Nobody has mapped it yet.

The four advantages are from Ethan Mollick, “The Overhang,” *One Useful Thing*, 18 September 2026.

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The Challenge



AI makes thinking scalable and **more important** than ever



AI misuse **weakens** your capacity to think, learn, and be creative

The goal: use AI to **raise the quality of your work while strengthening your own capacity** to think, learn, and create.

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Roadmap

- Why now: knowledge matters more
- From assistant to agent, with Exercise 1
- 15-minute break
- Put your agent to work, with Exercise 2
- Outsource the work, not the thinking
- Takeaways



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I have one main goal with this lecture: to help you learn how to work with **AI agents** to improve your **capacity to think** and your **ability to learn**!



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Important information



This is an inspirational lecture on thinking with AI. It discusses practical AI workflows, but it is not a general permission to use AI in your studies. Always follow the AI instructions, rules, and regulations for your course module. When in doubt, ask the course convener for exact guidance.

The views and workflows are my own. They build on public AI commentators, credited at the end.

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Get Ready While We Talk

Now, in the background **You will need this in Exercise 1. The download runs while we talk.**

1. Download the **ChatGPT desktop app** for Mac, Windows, or Linux. The link is on the lecture page.
2. Open it and **create a free account**, or log in. **Free is enough for today.**
3. Download the ready-made `ai-lab` **folder** and unzip it into your **Documents** folder.



Open on your laptop

timgasste.github.io/ai-thinking

Download links, prompts, the starter folder, and these slides

Stuck, or just curious? Ask at any point. There are no stupid questions about AI.

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Quick Poll: How Do You Use AI Today?

2 minutes, on your phone While the app downloads on your laptop. Anonymous.



1. Take out your **phone.**
2. **Scan the code,** or open **menti.com/al7tbh775af3**
3. Answer **four quick statements.**

Results appear live on the next slide.

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Quick Poll: Live Results

Join the poll
menti.com/al7tbh775af3

Live results appear here during the lecture

Scan the code on the previous slide to take part.

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From Assistant to Agent



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AI Tool Map (Click Tools to Explore)

General assistants
Reasoning, drafting, comparison, feedback, and broad task support.

Search and sources
Find current information and check answers against sources.

Literature
Discovery, citation context, and evidence assessment.

Writing and references
Language polish and reference management.

Coding agents
Work in project folders: files, code, analysis, and tests.

Transcription and voice
Speech capture, transcription, dictation, and generated audio.

OpenAI ChatGPT + Codex + Whisper	**Anthropic** Claude + Claude Code	**Gemini** Workspace + Code Assist	**Grok** Grok Build	**Microsoft Copilot**	**Perplexity**
NotebookLM now Gemini Notebook	**Elicit**	**Semantic Scholar**	**Scite**	**Litmaps**	**Zotero**
Grammarly	**GitHub Copilot**	**Cursor**	**Otter**	**Wispr Flow**	**ElevenLabs**

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From Prompt Engineering to Question Engineering



Then: prompt engineering

Templates, step-by-step chains, and long rule lists that walked the model through the task.



Now: question engineering

Frame a great question: a clear goal, real context, and your standard of quality. Then let the model plan the steps.

You do not need intricate prompt strategies. You need great questions. Strong models plan better than our scripts, and over-specifying imposes our own, often flawed, framing of the problem.

Then/now contrast based on Ethan Mollick, “The Dot and the Swarm,” *One Useful Thing*, 1 October 2026.

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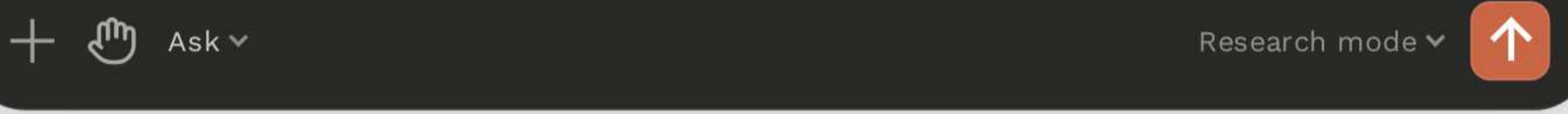
Anatomy of a Good Question



1. **Task**
2. **Purpose**
3. **Context**
4. **Material**
5. **Constraints**
6. **Format**
7. **Evaluation**
8. **Interaction**

Written out in full. Most questions need only a few of these.

(1) Help me assess whether political exclusion increases the risk of organized political violence. **(2)** I want to sharpen a research question before drafting a theory section, not receive a finished argument. **(3)** I am working in peace and conflict research and preparing a graduate seminar memo, so focus on mechanisms, scope conditions, alternative explanations, and observable implications. **(4)** Use only the notes and source excerpts I paste below. **(5)** Separate author claims from your inferences, mark unsupported claims as "needs source", and do not invent citations. **(6)** Return a four-step causal pathway, two counterarguments, three questions I must answer, and one next prompt. **(7)** A strong answer is specific, source-bounded, mechanism-focused, and honest about uncertainty. **(8)** Ask one clarifying question if needed, then wait for my response before revising the research question.

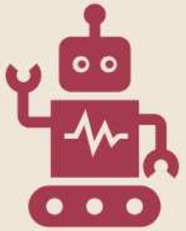


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Vibe Coding or Agentic Engineering?



Andrej Karpathy



Vibe coding

AI leads. The human steers by feel, accepts plausible output, and may miss errors, structure, and consequences.



Agentic engineering

AI executes. The human reviews, tests, and remains the architect, evaluator, and accountable author.

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The Basics of Agentic Engineering I



Model

A large language model that turns what you give it into an answer. Reasoning models think longer and check their own work.



Agent

A model working in a loop. It uses tools and files, checks the results, and keeps going until the task is done.



Harness

Everything around the model that keeps it on track, including instructions, files, tools, permissions, and memory.



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The Basics of Agentic Engineering II



Conventional model

Core language model

Examples: ChatGPT Instant and Claude Haiku 5.5



Reasoning model

Thinks longer before it answers

Examples: GPT-6 at high effort and Claude Opus 5.5



Reasoning model in an agent harness

Model plus tools, files, memory, and feedback

Examples: ChatGPT Work, Codex, or Claude Code in a local project folder with clear instructions, relevant files, tools, saved outputs, and human review

Adapted from Sebastian Raschka's blog post and figure.

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From Chatbot to Agent

One of the greatest leaps in how we can use AI: deploy your agent where the work lives.

Chatbot

The way most people use AI

- Answers when you paste material in.
- You move text, files, and results in and out.
- Best for dialogue, critique, explanation, and quick drafting.

Agent

The most powerful way to use AI

- Works inside a chosen project folder.
- Can read your readings and notes, code interview transcripts, run statistics, and save outputs.
- Best for multi-step research tasks that need files, tools, and review.

Build Your Harness

Start with a bounded local project folder. Add clear instructions, relevant files, tool permissions, and saved outputs. You remain the director. Define the goal, guide execution, and decide what gets used.

Choose a Tool

Use a tool that fits your purpose.

It must support agentic work.

It should run in a local project folder.

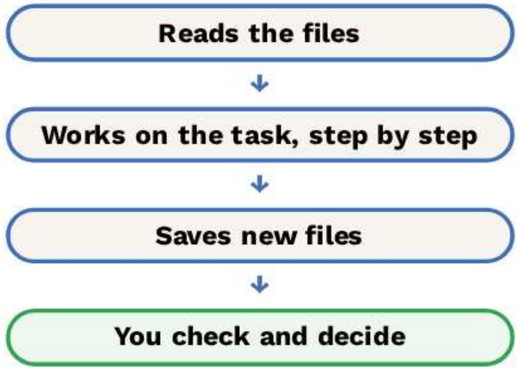
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Basic Setup: Folder, Agent, Instructions



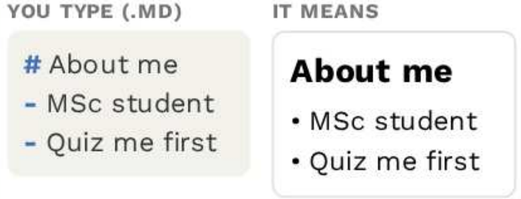
1 A folder on your computer

An ordinary folder in **Finder or File Explorer**, not something inside the AI app. You choose which folder the agent may open.



2 The app deploys an agent

Create a project from that folder, and the app does more than chat. Its agent reads, works, and saves there. You stay the director.



3 Instructions in a text file

Markdown (`.md`) is plain text with a few marks: # for a heading, - for a bullet. Models read it most reliably.

Prefer Word to Markdown, or Excel to `.csv`? That is completely fine. What matters is that the model gets your information as text and numbers, in a form that fits you and your workflow.

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Setting It Up and Using It Responsibly

One project folder. One tool connected to it. One thinking-oriented loop.

Set up the working folder

```
project/
|-- sources/
|-- working/
|-- outputs/
|-- LOG.md
+-- INSTRUCTIONS.md
```

Use one standing instructions file, such as `INSTRUCTIONS.md` (a Word file works too). Give access to this folder, not your whole computer.

Run it where the work lives

Use the desktop app rather than only the browser chat. On large jobs, ask for the plan before it runs.

Apply a thinking-oriented workflow

Plan → Prompt → Evaluate
Revise → Produce → Learn

Tool choice

Start free with the ChatGPT desktop app and its Work mode. For more, pick one paid stack and learn it well.

OpenAI — ChatGPT + Codex

Anthropic — Claude + Claude Code

Confidentiality and Copyright

Do not send confidential data or copyrighted material without approval or clearance.

File access

The tool can edit files. Keep sources/ read-only. Require approval before overwriting files or making larger revisions.

Verification

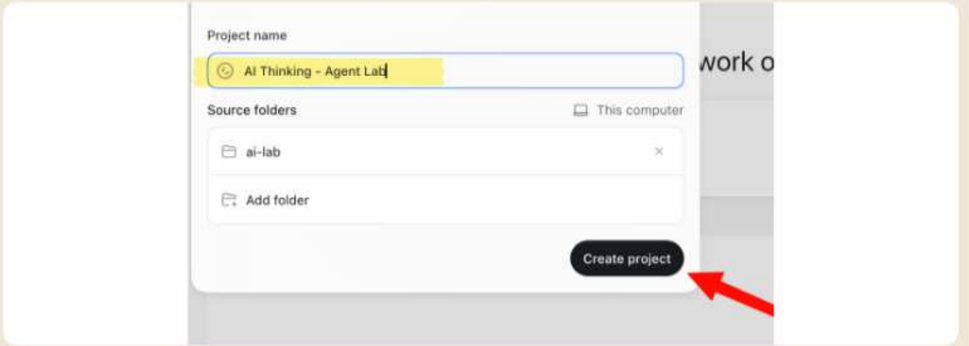
Instruct the tool to double-check claims and citations and never invent sources. You remain responsible. Check every number, quote, and source yourself.

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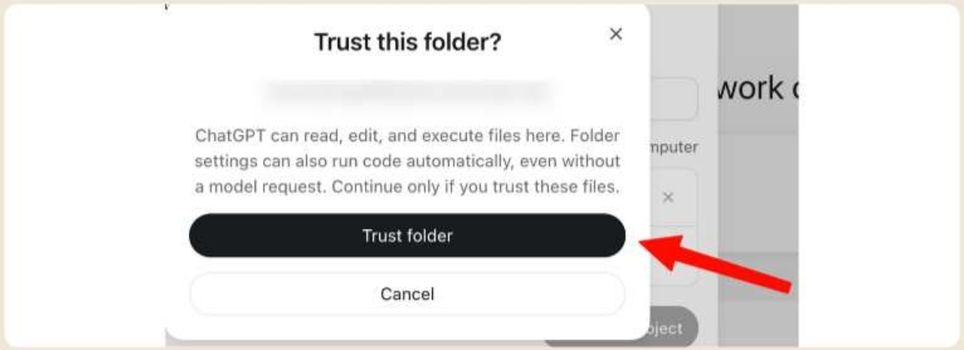
Exercise Preparation: Open Your Folder in ChatGPT



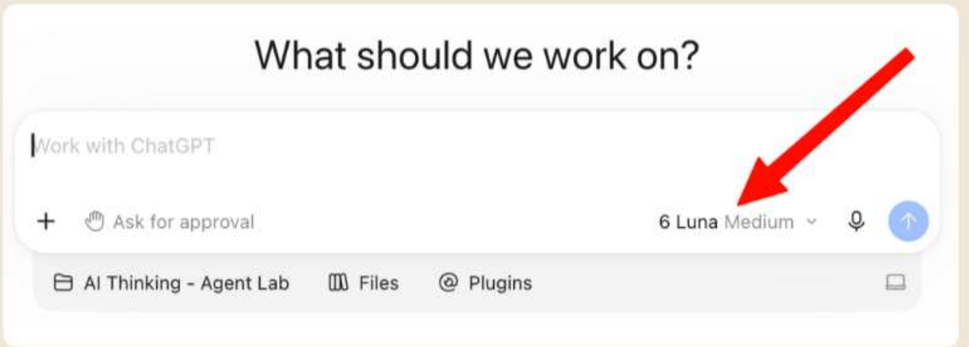
1 **Choose Work at the top, then click the + next to Projects.**



2 **Click Add folder, pick ai-lab in Documents, and click Open. Give the project any name at the top, then click Create project.**



3 **Click Trust folder. The agent can now read and save files in the ai-lab folder on your computer, and nowhere else.**



4 **Ready. Keep Ask for approval on and the default model. Then paste your first task.**

These steps are also on the lecture page: timgasste.github.io/ai-thinking

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Prompts and help, on your laptop

timgasste.github.io/ai-thinking

Exercise 1: Set Up Your Local Agent!

10 minutes **Build the harness you will use for the rest of the lecture. Help your neighbor if you finish early.**

1. **Find the `ai-lab` folder in your Documents folder. No download? Create a normal folder called `ai-lab` there, with subfolders `sources` and `outputs`.**
2. **In the ChatGPT desktop app, choose Work and create a project from `ai-lab`, as on the last slide. Another app whose agent can work in a folder works too.**
3. **Give it the task on the right, and answer its five questions. Short answers are fine.**
4. **Open `INSTRUCTIONS.md` (or `.docx`). Check it, and change one thing yourself.**

YOUR FIRST TASK

> Interview me with five short questions, one at a time, about my studies, my goals, and how I want you to help me learn. Then save my answers and these rules as `INSTRUCTIONS.md` in this folder: use only files in this folder, never invent sources, and ask before you overwrite a file.
>
> \+ Work · ai-lab

Prefer Word? Ask for `INSTRUCTIONS.docx` instead (about a minute slower). Markdown (`.md`) is plain text, which models read most reliably.

```
ai-lab/
|-- INSTRUCTIONS.md ← your harness
|-- sources/
+-- outputs/
```

Struggling to get the agent working on your computer? Pair up with a neighbor, or paste the task into a normal chat and save the answer as `INSTRUCTIONS.md` (or `.docx`) yourself.

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15-minute break

Questions are welcome before you go, and there are no stupid ones.



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Put Your Agent to Work



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Agentic Engineering: Summary



Powerful and accessible

Today we use a free, lighter model. With a stronger model and a creative agent setup, the sky is the limit.



Model + harness + context

The human guides the work, and the harness gives the model context, tools, and feedback. They turn a powerful motor into a race car with a driver.



Responsibility scales too

As agentic setups gain more access, security and ethics matter even more. Make them part of the setup from the start.



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Now let’s put agentic engineering to work: **to think better and learn more.**



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Example I: Your Study Partner

Turn cleared course material into active practice that fits you.

1 FOLDER

```
study-partner/
|-- INSTRUCTIONS.md
|-- sources/
    +-- cleared readings, notes, slides
        (PDF, Word, or text)
+-- outputs/
    |-- flashcards.csv
    |-- quiz.md
    +-- weak-spots.md
```

2 ASK IT FOR

- **Socratic tutoring**
- **Explanations at your level**
- **Flashcards for your gaps**
- **Audio scripts to listen to**
- **Voice: talk it through**
- **A scheduled weekly digest**

3 PROMPT

> Read my INSTRUCTIONS file. Using only `sources/`, quiz me on the hardest concept, one question at a time, and wait for my answer before you explain. Each time I get one wrong, add a flashcard to `outputs/` right away.

+ Work · study-partner

You still do the learning. Answer before you look, and check every card against the source.

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Example II: Your Sparring Partner

Challenge your argument before it becomes text.

1 FOLDER

```
argument-review/
|-- INSTRUCTIONS.md
|-- sources/
|-- working/
    +-- outline.md
+-- outputs/
    |-- steelman.md
    |-- weak-links.md
    +-- revision-log.md
```

2 ASK IT TO

- **Steelman the other side**
- **Find unsupported leaps**
- **Expose vague concepts**
- **Predict reader pushback**

3 PROMPT

Read my INSTRUCTIONS file and `working/outline.md`. Do not rewrite my text. Save a steelman of the strongest objection and a list of my unsupported leaps to `outputs/` first. Then ask me three hard questions.

\+ Work · argument-review

The AI critiques. You revise and write. Rerun the critique after each revision.

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Prompts and help, on your laptop

timgasste.github.io/ai-thinking

Exercise 2: Put Your Agent to Work

10 minutes Use this lecture's PDF in `sources/`, or add your own notes (Word files are fine). Then pick one option. One is enough.

A Study partner

Practice retrieval on something you need to know.

> Read my INSTRUCTIONS file. Quiz me on **[concept]** using only `sources/`, three questions, one at a time. Each time I get one wrong, add a flashcard to `outputs/` right away.

Saves `outputs/flashcards.csv`

B Sparring partner

Stress-test an argument you are working on.

> Read my INSTRUCTIONS file. Steelman the strongest objection to **[my argument]** and list my unsupported leaps. Do not rewrite my text. Save it to `outputs/` first, then show me.

Saves `outputs/steelman.md`

C Explain and listen

Get a hard idea explained the way you learn best.

> Read my INSTRUCTIONS file. Explain **[hard concept]** from `sources/` at my level, as a three-minute script I can listen to. Save it to `outputs/` first, then show me.

Saves `outputs/explainer.md`

Done early? Try another option. **Afterwards:** tell your neighbor one thing the agent got wrong, or one thing you now understand better.

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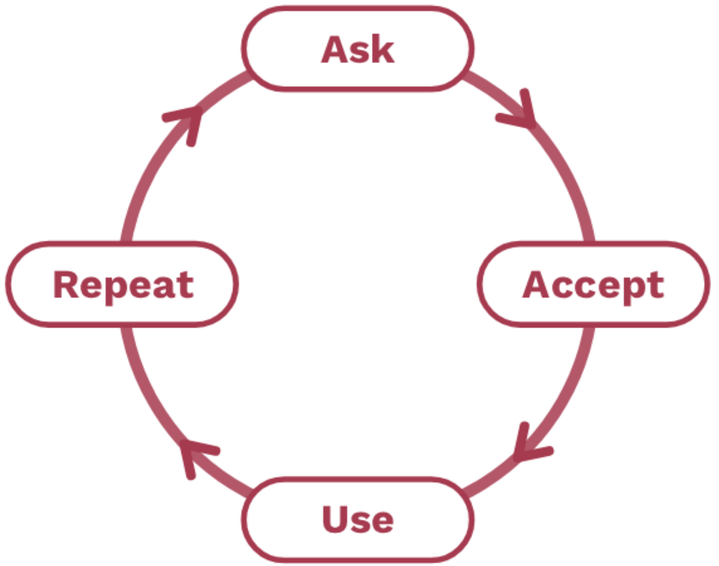
Outsource the Work, Not the Thinking



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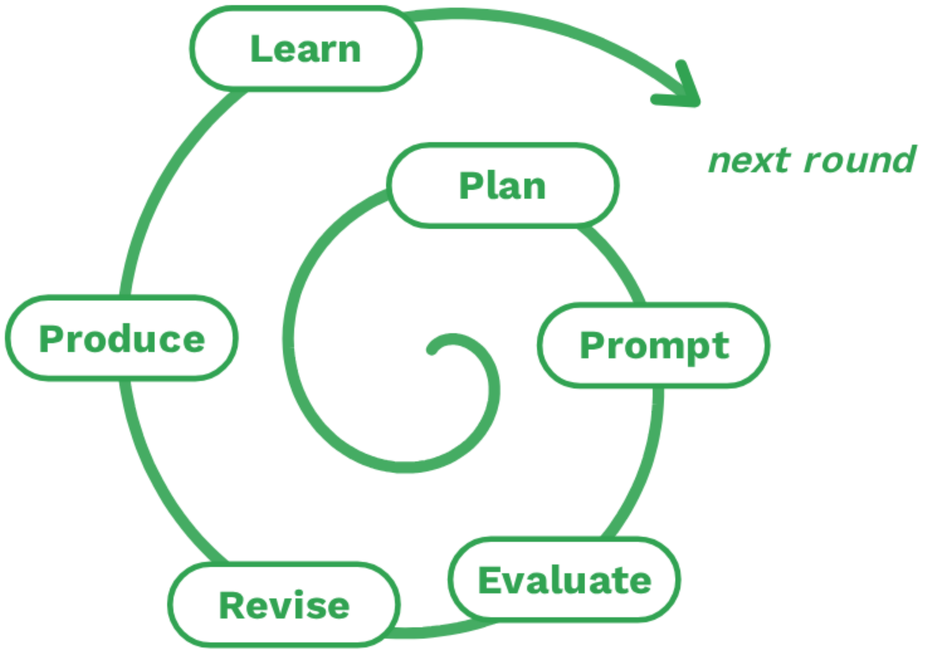
A Loop or a Spiral?

Output-oriented



Fast, but you end where you started.

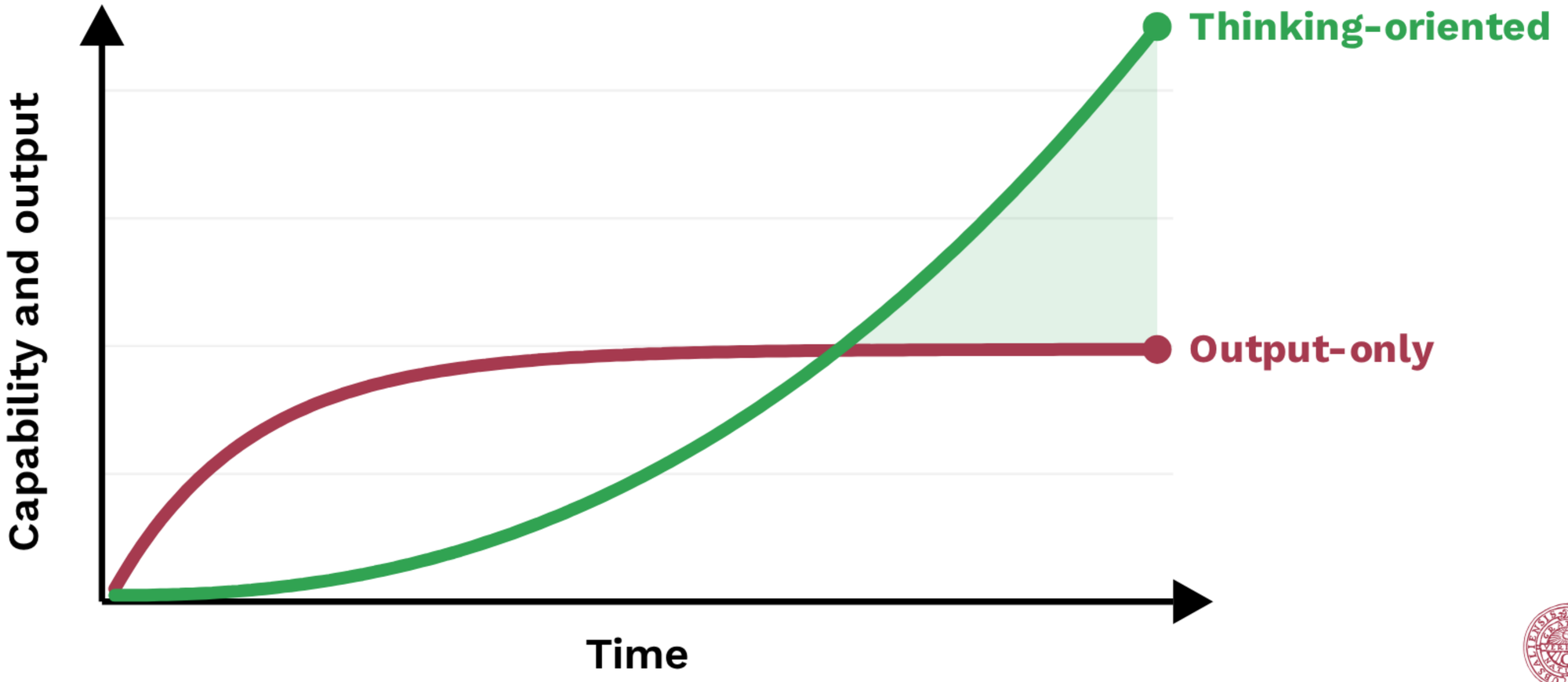
Thinking-oriented



Slower, but each round starts higher.

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Thinking Compounds



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Ultimately, Make the Knowledge Yours!

- Can I explain each claim and the reasoning behind it?
- Are the claims underpinned by credible evidence?
- Do I fully understand the method I am using?
- Can I defend all the choices in the work I submit?
- **Am I developing my own capacity to think, learn, and be productive, and am I using AI to amplify it?**

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Takeaways

1. AI moves the hard thinking upstream. **Knowledge, taste, and judgment matter more, not less.**
2. **Your capacity to think is the bottleneck.** Build deep and wide knowledge, and explore with agency.
3. **Start with one general assistant**, learn it well, and frame great questions. That is the most value for the least effort.
4. **Agentic engineering** puts the model in a harness: one folder, clear instructions, and your review. **Start free today.**
5. **Outsource the work, not the thinking.** Thinking-oriented use is a spiral, and it compounds.
6. **Make the knowledge yours.** If you cannot explain and defend it, it is not yours yet.

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Built On

The views and workflows in this lecture are my own. Many of the ideas build on these commentators, and I recommend following them.

Why knowledge matters

Ethan Mollick (2026). "The Overhang." *One Useful Thing*, 18 September. oneusefulthing.org/p/the-overhang

Ethan Mollick (2026). "The Dot and the Swarm." *One Useful Thing*, 1 October. oneusefulthing.org/p/the-dot-and-the-swarm

Agentic engineering

Andrej Karpathy coined "vibe coding" (February 2025) and proposed "agentic engineering" (February 2026) in posts on X.

Sebastian Raschka (2026). "Components of a Coding Agent." *Ahead of AI*, 4 April. magazine.sebastianraschka.com

Quotes, news, and listening

Jensen Huang (2025). Remarks at the Milken Institute Global Conference, 6 May. Reported by CNBC, 28 May.

NPR (2026). "Why AI companies are hiring philosophers to help develop their models." *All Things Considered*, 7 July.

Jonas von Essen and Benjamin Verbeek. *AI med Jonas & Benjamin* (podcast, in Swedish). Jonas von Essen on YouTube: youtube.com/@BlueberryClubSparven

Image credit: Jensen Huang, official White House photo by Daniel Torok (public domain), cropped.

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Questions?

1. **Ask out loud**, any question.
2. Rather not ask in front of everyone? **Scan the code** or open **menti.com/al446n8s1z42** and type it there.
3. Please share **two minutes of feedback.**

Answers appear live on the next slide.

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Your Questions and Feedback

Questions and feedback
menti.com/al446n8s1z42

Live answers appear here during the lecture

Scan the code on the previous slide to take part.

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Thank you!

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Slides, prompts, and links: timgasste.github.io/ai-thinking



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